

FinShield UPI Fraud Transaction Business Problem

1. Business Context

A digital payments business wants to analyze UPI transactions to identify fraud patterns, suspicious transaction behavior, and areas where fraud monitoring can be improved.

2. Business Problem

Identify the major patterns and factors associated with fraudulent UPI transactions and determine which transaction, customer, device, location, and risk-related characteristics require closer monitoring.

3. Business Objective

Measure the scale of fraud, understand fraud patterns, identify high-risk transaction segments, and provide data-driven insights that can support fraud detection and risk-monitoring decisions.

4. Key Business Questions

1. What percentage of transactions are fraudulent?
2. How much financial value is associated with fraudulent transactions?
3. Which transaction types have higher fraud occurrence?
4. Which payment methods or channels show suspicious patterns?
5. Does transaction amount differ between fraudulent and non-fraudulent transactions?
6. Which locations show higher fraud activity?
7. Are unusual location changes associated with fraud?
8. Which device or user behavior patterns are associated with fraud?
9. How does IP risk score relate to fraudulent transactions?
10. Which time periods have higher fraud activity?
11. Which transactions should be considered high-risk for monitoring?

5. KPIs / Metrics

- Total Transactions
- Fraudulent Transactions
- Fraud Rate %
- Total Transaction Amount
- Fraudulent Transaction Amount

- Average Transaction Amount
- Average Fraudulent Transaction Amount
- High-Risk Transaction Count
- Fraud Rate by Transaction Type
- Fraud Rate by Location / Segment

6. Data Required

The analysis should use the relevant columns available in the uploaded dataset, including:

- Transaction_ID
- Transaction_Date
- Sender_ID
- Receiver_ID
- Sender_Bank
- Receiver_Bank
- City
- Device_Type
- Transaction_Type
- Amount_INR
- Transaction_Status
- Account_Age_Days
- Transactions_Last_24H
- Avg_Transaction_Amount_7D
- Failed_Transactions_24H
- IP_Risk_Score
- Device_Risk_Score
- Location_Change_KM
- Fraud_Reason
- Fraud_Label
- Is_Fraud

7. Data Quality / Cleaning

Before analysis, validate data types, missing values, duplicate transactions, spelling/format consistency, invalid ranges, and suspicious values. Business-rule checks should be performed before removing records.

8. Analysis Approach

Use the following practical workflow:

- Python / Pandas --- initial cleaning and validation and EDA and deeper pattern investigation
- SQL — KPI calculation, segmentation, and fraud-pattern analysis
- Power BI — KPI monitoring and interactive fraud-risk dashboard

9. Findings

Document factual observations from the analysis, such as fraud-rate differences across transaction types, locations, amounts, risk scores, or behavioral segments. Do not label a variable as a cause of fraud unless the analysis supports that conclusion.

10. Root Cause / Driver

Investigate the strongest observed drivers or patterns behind fraudulent transactions. Separate direct observations from hypotheses that require further investigation.

11. Business Impact

Translate fraud patterns into business impact using transaction value, fraud volume, customer impact, operational risk, and potential financial exposure where the dataset supports it.

12. Recommendation

Recommend evidence-based monitoring or investigation actions for the segments showing higher fraud risk. Recommendations should be linked directly to the findings.

13. Monitoring

Create a recurring dashboard/report to monitor Fraud Rate %, Fraudulent Transaction Amount, high-risk transactions, fraud by transaction type, location, time period, and other important segments.

Note

This document defines the business-analysis structure from the uploaded dataset. It does not claim that the uploaded data represents actual RBI/NPCI production transactions.